

David Nguyen

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Embedded & Mobile Systems

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PROFILE

Senior software engineer with 10 years in industry plus a Ph.D. in mobile systems. At **CARIAD (Volkswagen Group)** I led the in-car App Store self-updates that reached **~60,000 production vehicles**. I also came up with the **AI-personalised App Store**, which set the store's future direction. I now work on the **Android Automotive OS** platform: the C++ Vehicle HAL, vehicle power management, and field defects that cross several systems. Before that I spent 6.5 years at **Meta** as an Android tech lead for Facebook Stories and Video, used by more than a billion people every day, and then worked at two Bay Area **startups**. I build and publish my own apps too: **CardioClowns** on the App Store (iPhone and Apple Watch, cardioclowns.com) and **StoreBench** on Google Play. 20+ publications, 6 granted US patents. Fluent in English, Czech and Vietnamese. I work across the stack, from Kotlin UI to C++ and the kernel, and I'm the person teams bring the hard bugs to.

EXPERIENCE

Senior Software Engineer · CARIAD SE (Volkswagen Group) · Ingolstadt 04/2024 – present
In-vehicle software for all VW Group brands (Audi, VW, Škoda, SEAT/CUPRA, Porsche, Bentley) on Android Automotive OS

Android Platform: Vehicle HAL & Power Management · 06/2026 – present

- Develop the **Vehicle HAL (VHAL)** and **Vehicle Power Service** in C++, which connect Android to the car's ECUs (power states, suspend-to-RAM, climate, input), together with Android's **CarService** in Java.
- Audited the **~1,300-file VHAL/VPS codebase** for reliability. Fixed whole classes of latent crashes (use-after-free, null dereference, unchecked optionals, uninitialised members) in small, low-risk PR batches before they could reach the field.
- Own root cause and fix for field defects reported from customer and test vehicles: **black screen after sleep** (a lost cancel in the power handshake between Android and the vehicle, fixed on 3 release branches), **no phone-call audio** (audio focus stall), **driver-distraction blocking** (relaunch storm cut 5–15×, 100 % blocking on VW and Audi benches) and **climate controls** dead at LO/HI.

Group App Store (in-car app store) · 04/2024 – 06/2026

- **Lead and main developer for App Store self-updates**, shipping new features and fixes over the air to **~60,000 production vehicles** already on the road. The Audi self-updates were presented as a success story at the organisation's town hall. For Audi Asterix I set up the release branch and CI/CD, planned the rollout and submitted the release to production (02/2026). I also ramped up developers on the CL46/CL51 and Porsche releases.
- **Came up with and led the AI-personalised App Store**, taking it from the first idea to the end-to-end architecture and a team presentation to the **CARIAD CEO**. It set the **new direction for the future App Store** and led to a joint CARIAD–Harman task force.
- **Technical lead of CARIAD's in-house store development** (2024), setting the quality bar for both in-house and Harman supplier code: hundreds of code reviews (10–31 a week). Brands (Audi, Porsche, VW) came to me directly with issues. Wrote the Hidden Features engineering-menu concept.
- **Release manager** on rotation for multi-brand deliveries, e.g. one CL53/CL48 release integrated for **7 brands** (Audi, VW, Škoda, CUPRA, Porsche, Bentley, Ford) plus MIB4 packages: builds, cherry-picks, smoke tests, handover to integration.
- Implemented **EU Digital Services Act (DSA)** compliance in the store client for all brands: age ratings, in-app-purchase, advertising and publisher information, and the "report illegal content" flow.
- Go-to engineer for hard cross-system defects between the store, Harman's Ignite SDK, HMI and backend (multi-user account switching, OEM app visibility, a 14-build bisection). Closed ~70 tickets and helped reach near-zero bugs for CL53.
- Interviewed ~17 candidates (two joined the team), onboarded and taught new engineers, and co-presented ["Developing Android Automotive Apps for the VW Group" at droidcon Berlin 2024](#).

- Software Engineer (Android Lead) · Mylio Photos** · San Francisco Bay Area 08/2023 – 04/2024
- Led Android development of a cross-platform photo-organisation app at a 29-person startup (9 engineers).
- Software Engineer · Mission Peak** · San Francisco 02/2023 – 08/2023
- Built Android compatibility-testing solutions for device OEMs at an early-stage startup.
- Engineering Lead (2018–2023) · Software Engineer (2016–2018) · Meta** · Menlo Park, CA 08/2016 – 01/2023
- Android tech lead for Facebook Stories and Video Player, used by more than 1 billion people every day*
- **Player Plugin Platform:** co-designed the Android infrastructure that lets ~20 product teams add features to the video player. The UI is driven from the server, so changes reach production in 1 day instead of 2 weeks.
 - **Stories pagination:** led the infrastructure that loads more stories in feed units, which had the largest engagement impact in a ~200-engineer org.
 - **Memory optimisation and connected home:** reduced app crashes by 14 % and launch delays by 78 % through heap-limit tuning. Built the hub-to-device BLE pairing, authentication and command app for a smart-home system. This work led to 5 granted US patents.
 - **Fire TV:** co-built the first Facebook Video app for Amazon Fire TV, bringing the product to about 50 M more users.
 - One of ~15 company-wide **Android release sheriffs**, gatekeeping Facebook app hotfixes. Managed 8 interns, all of whom got full-time offers (Meta, Google, ...). Mentored engineers and ran 100+ interviews.
- Research Software Engineer / Ph.D. Candidate · William & Mary** · Williamsburg, VA 09/2011 – 07/2016
- **SmartIO** (ACM MobiSys 2015): large-scale study of Android I/O delay on 2,611 devices. Prioritising reads over writes in the I/O stack cut app launch delay by up to 38 % and runtime delay by up to 30 %. Led to multiple publications and a startup.
 - **SmartStorage:** first study of how storage affects smartphone energy use, down to the block layer and device driver. Dynamically tuning storage parameters saved 23–52 % energy across the 20 most popular apps. Led to a [granted US patent](#).
- Engineering Intern (2×) · Facebook** · Menlo Park, CA 05/2014 – 08/2014 · 06/2015 – 09/2015
- First hub-to-device app for what became Facebook Portal (2015). Memory optimisations for the Facebook Phone project (2014).
- Lecturer · Christopher Newport University** (2013) · **Suffolk University** (2009–2011) 09/2009 – 05/2013
- Taught data communication systems, programming, web development (HTML/CSS/JavaScript/PHP) and calculus.

EDUCATION

- Ph.D., Computer Science** · William & Mary, USA 2011 – 2016
- Mobile systems, storage and performance · [20+ publications](#) (ACM MobiSys, UbiComp, IEEE INFOCOM, IEEE TMM, ...), 360+ citations, h-index 10 · 6 granted US patents ([5 at Meta](#), [1 at William & Mary](#))
- M.Sc., Computer Science** · Suffolk University, Boston, USA 2010
- Visiting Graduate Student**, Algorithms & Data Structures · Stanford University, USA 2009
- Bachelor, Mathematics & Computer Science** · Charles University, Prague, Czech Republic 2007

SKILLS

Programming	Kotlin, Java, C++, Swift (iPhone, Apple Watch), Python, Shell; JavaScript, React Native
Automotive	Android Automotive OS (AAOS), Vehicle HAL / AIDL, CarService, vehicle power management, over-the-air app updates, multi-brand release & integration, Harman Ignite SDK, test benches and car tests, multi-ECU trace analysis
Android	Multi-display / multi-user apps, Gradle/AGP & JDK upgrades, Retrofit/OkHttp, Robolectric, AOSP (Soong), Binder, SELinux, Linux kernel & storage/memory performance, Bluetooth LE
Engineering	GoogleTest, JUnit, hardware-in-the-loop testing, GitHub Actions, Docker, Git, Gerrit, Jira; code review, mentoring, hiring
Spoken languages	English (native) · Czech (fluent) · Vietnamese (fluent) · German (communicative)
Nationality	US citizen · EU permanent resident, no visa sponsorship required